MD Khalequzzaman Chowdhury Sayem

Pre-Doctoral Researcher, STARS Team, Inria, Sophia Antipolis, France

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Research Summary

Pre-Doctoral Researcher in the **STARS team at Inria**, working on multimodal foundation models and human-centric video understanding. Previously, a 3D AI Engineer at **LiOPS** (3D perception for industrial robotics). At UNIST, I worked on real-time two-hand manipulation modeling (**AAAI 2025**), **HandVQA** (**CVPR 2026**), a **1.6M**-scale benchmark for spatial reasoning about hands in VLMs, and 3D Gaussian human avatars (**ECCV 2026**). I am interested in models that understand people: how they move, interact with objects, and behave over time.

Research Interests

Human Behavior Understanding; Multimodal Foundation Models; Video Understanding; 3D Computer Vision; Human-Object Interaction; Vision-Language Models; World Models; Embodied AI.

Education

M.S. in Computer Science and Engineering — **Sep 2023 – Aug 2025**

Ulsan National Institute of Science & Technology (UNIST) — *Ulsan, Korea*

- GPA: 3.93/4.3 (96.30/100)
- **Master's Thesis (Accepted at CVPR 2026):** *HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.*
- Advisors: Prof. Seungryul Baek (Supervisor), Prof. Binod Bhattarai (Co-supervisor)

B.S. in Industrial Engineering — **Sep 2019 – Aug 2023**

Ulsan National Institute of Science & Technology (UNIST) — *Ulsan, Korea*

Conference Publications (* equal contribution)

- [**CVPR 2026**] **MD K. C. Sayem***, M. T. Chowdhury*, Y. Y. Tiruneh, M. A. Khan, M. S. Ali, B. Bhattarai, S. Baek.
  HandVQA: Diagnosing and Improving Fine-Grained Spatial Reasoning about Hands in Vision-Language Models.
  Project Page: *kcsayem.github.io/handvqa/*
  - Proposed **HandVQA**, a **1.6M-scale** benchmark derived deterministically from **3D hand pose annotations** across FreiHAND, InterHand2.6M, and FPHA.
  - Diagnosed systematic failures of large VLMs (e.g., LLaVA, Qwen2.5VL) in fine-grained spatial reasoning, revealing depth ambiguity, left/right confusion, and relational bias.
  - Demonstrated that injecting **structured 3D supervision** improves zero-shot transfer across tasks, boosting gesture recognition by **+10.33%** and hand–object interaction recognition by **+2.63%**.
  - Provides a scalable pathway for injecting explicit **3D geometric priors** into text-first foundation models.
- [**ECCV 2026**] Y. Y. Tiruneh, M. S. Ali, U. Jeong, M. A. Khan, **MD K. C. Sayem**, A. Bayramgeldiyev, B. Bhattarai, S. Baek.
  FlexiAvatar: Unified 3D Gaussian Human Avatars Under Arbitrary Body Visibility.
  Project Page: *yihalem1.github.io/FlexiAvatar/*
  - Visibility-aware framework for animatable **3D Gaussian human avatars** from full-body, upper-body, or head-only monocular video.
- [**AAAI 2025**] E. Ismayilzada*, **MD K. C. Sayem***, Y. Y. Tiruneh, M. T. Chowdhury, M. Boboev, S. Baek.
  QORT-Former: Query-optimized Real-time Transformer for Understanding Two Hands Manipulating Objects.
  Project Page: *kcsayem.github.io/QORT-Former/*
  - Designed the core query optimization method and integrated structured contact-map features into the Transformer architecture.
  - Proposed a **query division strategy** that separates queries into left-hand, right-hand, and object groups via a dedicated query proposal module, reducing ambiguity between symmetric hand features.
  - Reduced Transformer complexity from **300 queries / 6 decoders** to **108 queries / 1 decoder**, achieving **53.5 FPS** real-time inference while improving state-of-the-art accuracy by **+27.2%** (H2O) and **+10.4%** (FPHA).

Research Experience

STARS Team, Inria
Nov 2026 – Present
Pre-Doctoral Researcher — *Sophia Antipolis, France*
– **Multimodal foundation models** and **human-centric video understanding**, supervised by Dr. François Brémond.

Vision & Learning Lab, UNIST
Jan 2023 – Apr 2026
Researcher — *Ulsan, Korea*
Researcher (Sep 2025 – Apr 2026); Graduate Research Assistant (Sep 2023 – Aug 2025); Undergraduate Research Assistant (Jan 2023 – Aug 2023)
– Designed geometry-aware Transformer architectures for real-time 3D hand-object interaction modeling, integrating structured contact and pose representations (AAAI 2025).
– Led large-scale experimentation across FreiHAND, InterHand2.6M, and FPHA to analyze cross-view generalization, occlusion robustness, and representation bias in multimodal systems.
– Developed deterministic data generation and diagnostic evaluation pipelines for fine-grained spatial reasoning in VLMs (CVPR 2026).
– Contributed to 3D Gaussian human avatar reconstruction under partial body visibility (ECCV 2026).

Machine Learning & Vision Lab, UNIST
Sep 2021 – Feb 2022
Undergraduate Research Assistant — *Ulsan, Korea*
– Improved data augmentation and training for 2D human pose estimation on COCO-style benchmarks.

Industry Experience

LiOPS
May 2026 – Oct 2026
3D AI Engineer — *Seoul, Korea*
– Developed **Blender-based synthetic 3D data pipelines** for model training and evaluation.
– Worked on **point-cloud learning** (Point Transformer V3) and **stereo depth estimation** for 3D reconstruction.
– Built geometry-based perception for **robotic manipulation**, object localization, gripper alignment, and robotic welding.

Selected Industry Research Project

Worker Safety System for Industrial Environments Using Multi-Task YOLOv11 **Feb 2025 - Apr 2025**
Hyundai Motors & Vision and Learning Lab (UNIST) | *GitHub*
– Built a unified multi-task perception system for detection, instance segmentation, and human pose estimation.
– Integrated YOLOv11 detection, segmentation, and pose heads into a single pipeline, reducing inference latency by **35%**.

Technical Skills

- **Core:** PyTorch, CUDA, Distributed/Accelerated Training, Linux, Git, LaTeX
- **Multimodal / Foundation Models:** HuggingFace Transformers, LoRA fine-tuning, VLM evaluation (LLaVA, Qwen-VL)
- **3D Vision:** Pose estimation, multi-view geometry, human/hand modeling, stereo depth, point clouds and LiDAR, multi-camera calibration, NeRF / Gaussian Splatting, ONNX / TensorRT
- **Data & Tooling:** NumPy, Pandas, OpenCV, Blender (synthetic 3D data), experiment scripting and evaluation pipelines

Teaching Experience

- **Teaching Assistant:** Introduction to AI Programming in Python (Fall 23, Spring 24, Fall 24) and Discrete Mathematics (Fall 24) with Prof. Michael Jon Burrell; Advanced Computer Vision (Spring 24) with Prof. Seungryul Baek.

Leadership

PAIS: Partnership for AI and Society
Jan 2021 – Aug 2025
President — *UNIST*
– Founded and led an AI-safety discussion group with faculty mentorship, organizing talks on data privacy and responsible AI.

Honors & Awards

- **Graduate School Scholarship (Korean Government):** full tuition + stipend for M.S.
- **UNIST Dream Scholarship (UNIST):** full tuition + stipend for B.S.
- **AI Challengers Program (UNIST Innovative Education Center): $5K** funding for parallelized bandit algorithms.